

Elecard Stream Analyzer 2024

User Guide

User Guide Notices

Elecard Stream Analyzer 2024 (4.7) User Guide

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1. Introduction

1.1 Preface

Elecard Stream Analyzer is a powerful tool designed for syntax analysis of encoded media streams and presentation of the analysis log in a human readable form. Stream Analyzer operates with MPEG-1 Video/Audio, MPEG-2 Video/Audio, AAC, Dolby Digital Audio, MPEG-4, AVC/H.264, VC1, HEVC/H.265, VVC/H.266, AVS3, AV1 and VP9 contained in MPEG-1 SS, MPEG-2 PS, TS, MP4, AVI, FLV, IVF and MKV/WebM, HEIC files.

1.2 Describing Elecard Stream Analyzer

The following section defines the specifications and features of Elecard Stream Analyzer.

1.2.1 Supported Media Types

Elecard Stream Analyzer supports the following formats:

- MPEG-1 Video stream (ISO/IEC 11172-2);
- MPEG-2 Video stream (ISO/IEC 13818-2);
- AVC/H.264 Video stream (ISO/IEC 14496-10) and its Annex G (SVC) and Annex H (MVC);
- HEVC/H.265 Video stream (ISO/IEC 23008-2, MPEG-H Part 2);
- MV-HEVC (ISO/IEC 23008-2 Annex G);
- VVC/H.266 Video stream (ISO/IEC 23090-3);
- VP9 Bitstream Overview for Google VP9;
- AV1 Bitstream & Decoding Process Specification;
- AVS3 (Information Technology Intelligent Media Coding (T/AI 109.2 — 2021) by Industry Alliance Standard) MPEG-4 Video stream;
- VC-1 Video;
- MPEG-1/2 Audio Layer 1/2/3 (ISO/IEC 11172-3 and ISO/IEC 13818-3);
- Dolby Digital Audio (ATSC A-52);
- AAC (Advanced audio coding);
- LPCM (Linear Pulse Code Modulated Audio);
- SCTE-35 (ANSI/SCTE-35);
- DVB Subtitle (ETSI EN 300 743);
- MPEG-1 System Stream (MPEG-1, MPEG-1/2 Audio Layer 1/2/3, LPCM) (ISO/IEC 11172-1);
- MPEG-2 Transport Stream (MPEG-1/2, AVC, HEVC, MPEG-1/2 Audio Layer 1/2/3, LPCM, AC-3, AAC, SCTE-35, DVB Subtitle) (ISO/IEC 13818-1);
- MPEG-2 Program Stream (MPEG-1/2, AVC, HEVC, MPEG-1/2 Audio Layer 1/2/3, LPCM, AC-3, AAC) (ISO/IEC 13818-1);
- AVI file container (MPEG-1/2, AVC, HEVC, LPCM, AC-3, AAC);
- MP4 file container (MPEG-1/2, AVC, HEVC, VP9, AV1, VC-1, AC-3, AAC);
- MKV file container (MPEG-1/2, AVC, HEVC, VP9, AV1, MPEG-1/2 Audio Layer 1/2/3, AC-3, AAC);
- WebM file container based on MKV (VP9, AV1);
- IVF file container (VP9, AV1);
- FLV file container (AVC, HEVC, AAC, LPCM);

- HEIC file container.

1.2.2 Features

Elecard Stream Analyzer implements the following features:

- Selection of packets in a text;
- Selection of packets by PID and StreamID;
- Stream viewing in the hex, ascii and binary modes;
- Storing information about the stream and currently selected packets into a .CSV file;
- Search of elements by offset, PID and text;
- Transport stream error detection (ETSI TR 101 290) and analysis, error report generation;
- Interleaving analysis;
- Presentation of the interleaving between two streams as a diagram;
- Calculation of the overhead in transport, program, and system streams;
- Bandwidth information;
- Customization main control output;
- MP4, AVI, MKV/WebM, FLV, IVF, HEIC support;
- Cross platform (Windows, Mac, Linux version);
- Selection of the data portion for further saving;
- Dump elementary streams to file;
- Build graphics from parameters;
- Sharing comments between application instances and/or applications of Elecard StreamEye Studio set;
- Synchronization between applications of Elecard StreamEye Studio set (Binding mode);
- Console version.

1.3 Using this Guide

1.3.1 Purpose

This guide is intended to help the user utilize Elecard Stream Analyzer. It describes the Stream Analyzer GUI, settings and functions and provides instructions for Stream Analyzer use.

1.3.2 Topics Covered

Section 1: Introduction – provides a general overview of Stream Analyzer and describes the purpose of the document and its contents.

Section 2: Getting Started – describes how to install, uninstall, and run the Elecard Stream Analyzer program.

Section 3: Using Elecard Stream Analyzer – describes the Stream Analyzer GUI and provides instructions for comparing video streams and viewing the results of comparison.

1.4 System Requirements

1.4.1 Hardware Requirements

- 8 GB RAM for HD video (or higher).
- 16 GB RAM for UHD video (or higher).

1.4.2 Software Requirements

- macOS 10.13 High Sierra — Mac OS 14 Sonoma;

1.5 Licensing and Technical Support

By installing, copying, or otherwise using the software product or any updates, you agree to be bound by the terms of the "Elecard" End-User License Agreement ("EULA"). This EULA is a legal agreement between you (either an individual or a single entity) and Elecard for the "Elecard" software product(s) accompanying this EULA, which include(s) computer software and may include "online" or electronic documentation, associated media, and printed materials ("software product").

For technical support, please contact Elecard Technical Support Team: tsup@elecard.com.

For sales and licensing information contact Elecard Sales Department: sales@elecard.com.

1.6 Activating Elecard Stream Analyzer

1.6.1 Sentinel License Protection

Make sure that Sentinel License Manager software has been installed on your computer. License Manager is provided within the product installation pack for Windows OS and macOS and installed automatically.

License Manager is provided as a separate installation pack for Linux OS and should be installed manually.

Please note, macOS reinstallation will make the activated product license invalid.

To check that License Manager is properly installed, go to Sentinel Admin Control Center at <http://localhost:1947>. If you see your license information, the installation is successfully completed.

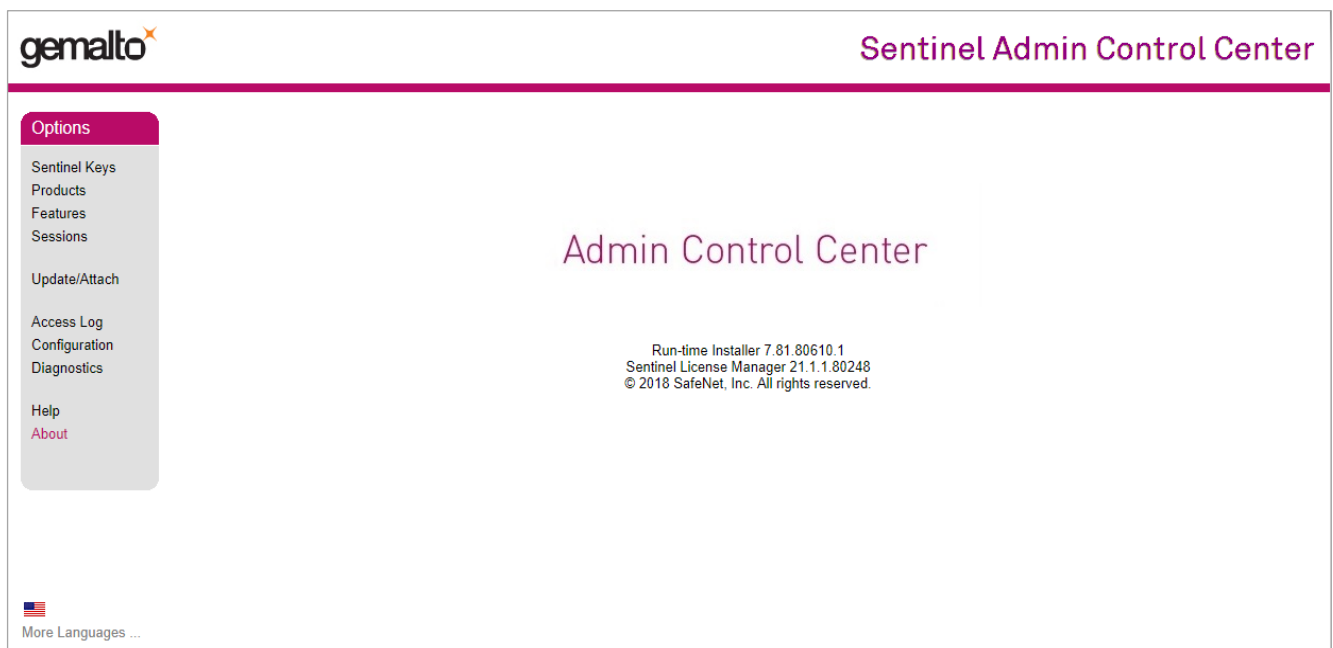


Figure 1. The License Manager with license information

License Manager allows configuring, controlling, monitoring and looking through a list of available licenses. For more details on License Manager operation click the Help tab.

Note, each page contains the *Help* tab related to this page only.

You can activate Elecard Stream Analyzer offline via following steps:

For Elecard Stream Analyzer activation, a C2V (client-to-vendor) file should be created. To create the C2V file follow the steps:

1. Open the **StreamAnalyzer4** tab and select **Preferences** in the drop-down menu.
2. A dialog window should open. Open the **License** tab.
3. Select HASP-SL for PC-based or HASP-HL for dongle-based activation in the *Offline activation with license files* field. Note, if you select a PC-based (HASP-SL) activation, full featured activated Elecard Stream Analyzer will run only on this particular hardware. If you have Elecard dongle, plug it into a free USB port on your computer and select dongle-based (HASP-HL) activation. It will be possible to use Elecard Stream Analyzer on any hardware with the dongle plugged in.

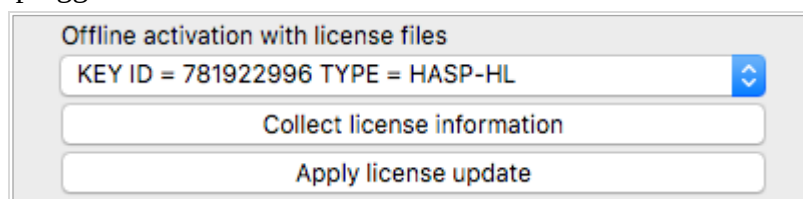


Figure 2.

Offline activation with license files

4. Click the **Collect license information** button. The Save File dialog box should open. Click **Save** and save the C2V file to a required directory. Send the saved C2V file to Elecard Sales at sales@elecard.com.

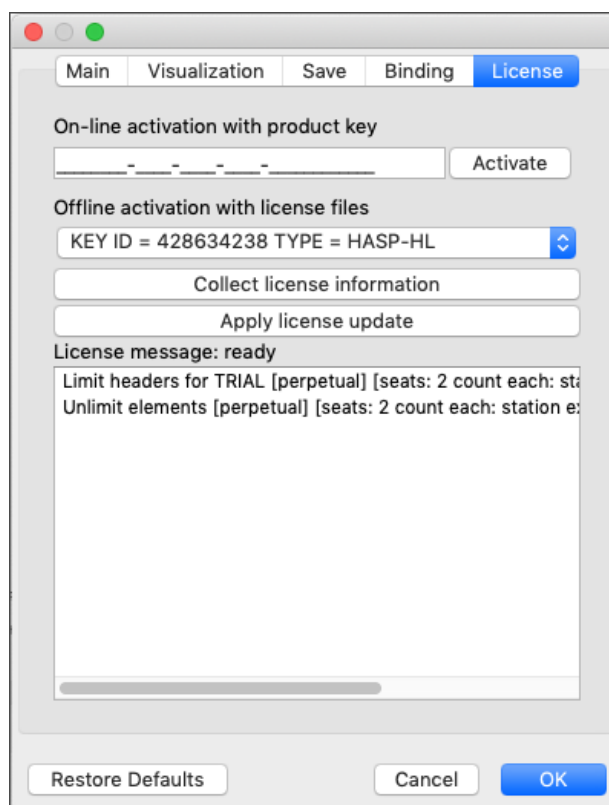


Figure 3.

Creating a C2V file: License tab

5. After receiving the V2C file containing license information back from Elecard Sales save the received file to a required directory. Open the **StreamAnalyzer4** tab and select **Preferences** in the drop-down menu again. Open the **License** tab and click **Apply license update** (see Figure

4). The file selection dialog box should appear. Choose the received V2C file and click **OK**.
Your application is successfully activated.

2. Getting Started

The following section details the procedures for installing and running Elecard Stream Analyzer.

2.1 Installing Stream Analyzer

Elecard Stream Analyzer is supplied as part of Elecard StreamEye Studio or as a standalone application. After installing Elecard StreamEye Studio or Elecard Stream Analyzer on macOS, all applications are located in the Applications folder:

- Applications → Elecard StreamEye Studio 2024;
- Applications → Elecard Stream Analyzer 2024.

The console is located in: Users → bin → SAconsole → streamanalyzer-cli.

The documents are located in: Library → Application Support → Elecard → Stream Analyzer 2024 → Elecard Stream Analyzer 2024.pdf.

2.2 Uninstalling Stream Analyzer

Elecard Stream Analyzer can be uninstalled as part of Elecard StreamEye Studio or as a standalone application. To uninstall the application in macOS, move the required application to the trash icon.

Elecard StreamEye Studio or Elecard StreamEye applications are located in the Applications folder:

- Applications → Elecard StreamEye Studio 2024;
- Applications → Elecard Stream Analyzer 2024.

The console is located in: Users → bin → SAconsole.

All documents are located in: Library → Application Support → Elecard.

2.3 Running Stream Analyzer

The installed Elecard StreamEye Studio or Elecard StreamEye including all files packaged together are located in *Applications → Elecard StreamEye Studio 2024* or *Applications → Elecard Stream Analyzer 2024*, and all consoles are located in *Users → bin → Elecard StreamEye Studio 2024*.

To run Elecard Stream Analyzer double click on the application icon, and the application will automatically run.

3. Using Elecard Stream Analyzer

3.1 Introduction

The following section describes the Elecard Stream Analyzer GUI (graphic user interface), its features and instructions for stream analysis. The Elecard Analyzer is an advanced video analysis software tool standing out for its simple user-friendly interface and in-depth statistics analysis.

3.2 Describing Elecard Stream Analyzer GUI

The following section describes the Elecard Stream Analyzer GUI.

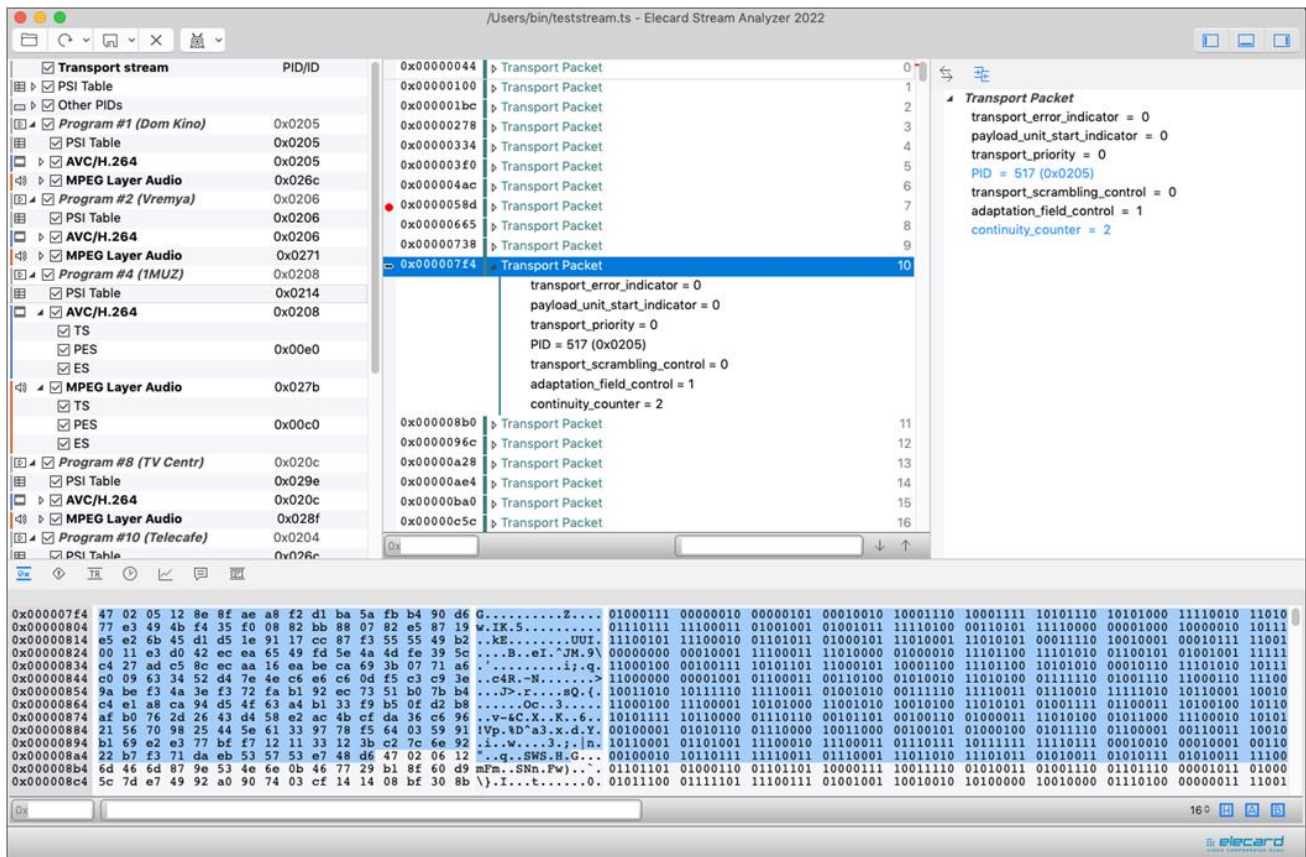


Figure 4. Elecard Stream Analyzer GUI

3.2.1 Menu

The following table describes the Elecard Stream Analyzer menu.

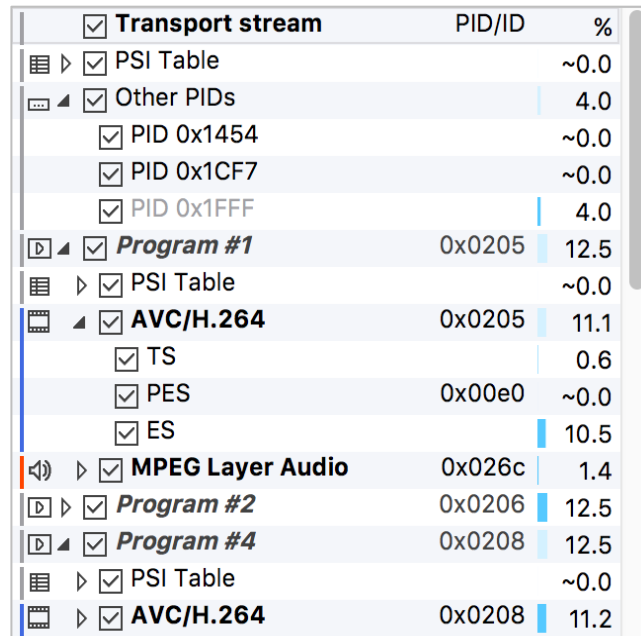
Table 1. Elecard Stream Analyzer GUI – Menu

Menu item	Button	Description
StreamAnalyzer4		
About (application)		Shows information about the current application and supported formats
Preferences		Opens the program settings dialog (Main, Binding, License).

Quit (application)		Closes the application.
File		
Open file		Opens a new media file for analysis.
Reopen		Reopens a list of recently opened files.
Reopen Last File		Reopens the last opened file.
Save		Allows selecting file information to be saved: - Stream Info – saves all information on a stream; - Dump – extracts and saves a piece of a file or an elementary stream and allows choosing the required data portion.
Close		Closes the opened file.
Binding		Enables or disables Master / Slave mode: - Leader – allows sending comments, commands on interface or positioning changes to other applications; - Follower – allows receiving comments, commands from other applications.
Left Panel		Hides/shows Explorer panel.
Bottom Panels		Hides/shows panels at the bottom of the application window: Hex Viewer, Messages, TR101-290, Time Dynamics, Graphics, Comments.
Right Panel		Hides/shows Property panel.
Open comments		Opens comments from the XML file.
Save comments		Saves comments to the XML file.
Open with		Opens the current file with an external application (the application name and its path are set in the Open With List field by opening Preferences – Main. Player is set by default).
Navigation		
Offset		Enables searching by the offset.
Search text box		Enables searching the necessary byte sequence in the file (helps to search for start-codes).
Previous / Next		Switches the searching results: previous / next.
View		
Property		Property panel containing the following options:
<i>Sync mode</i>		automatically changes data displayed in the Property panel
<i>Compare mode</i>		compares the chosen packet with the selected one.
<i>Dump mode</i>		saves an elementary stream selected in the Explorer panel.
Hex Viewer		Activates the Hex Viewer panel
Time Dynamics		Activates the Time Dynamics panel
Messages		Activates the Messages panel
TR101-290		Activates the TR101-290 panel
Graphics		Activates the Graphics panel

Comments		Activates the Comments panel
Help		
Contents		Opens the current User Guide in .pdf format
Feedback		Sends the Stream Analyzer feedback via e-mail, if it is registered in the system with the following subjects: <i>Submit an Idea, Ask a Question, Report a Bug.</i>
Search		Searching tools of the application.

3.2.2 Explorer Panel



☒	Transport stream	PID/ID	%
☒	PSI Table		~0.0
☒	Other PIDs		4.0
☒	PID 0x1454		~0.0
☒	PID 0x1CF7		~0.0
☒	PID 0x1FFF		4.0
☒	Program #1	0x0205	12.5
☒	PSI Table		~0.0
☒	AVC/H.264	0x0205	11.1
☒	TS		0.6
☒	PES	0x00e0	~0.0
☒	ES		10.5
☒	MPEG Layer Audio	0x026c	1.4
☒	Program #2	0x0206	12.5
☒	Program #4	0x0208	12.5
☒	PSI Table		~0.0
☒	AVC/H.264	0x0208	11.2

Figure 5. Elecard Stream Analyzer GUI – Explorer Panel – TS Stream

To open the **Explorer** panel, click **View – Explorer**.

This panel shows the stream hierarchical tree and allows managing information display in the **Main window** panel. Left-click on the element shows all properties in the **Property** panel. Right-click on the element shows the shortcut menu.

The **Percentage** field shows the percentage rate of an element in the whole stream.

When choosing some stream element, you can see two more functions in the **Property** panel.

The **Dump** button saves the dump file.

Use checkboxes to select data that you need displayed in the **Main Window** panel.

MP4 streams have descriptive headers that contain detailed information about MP4 boxes (atoms), which is displayed in the panel.

☒ MP4 stream	size	offset
ftyp	24	0x00000000
☒ moov	10 052	0x00000018
mvhd	108	0x00000020
☒ trak	9 936	0x0000008c
tkhd	92	0x00000094
mdia	6 884	0x000000f0
mdhd	32	0x000000f8
hdlr	32	0x00000118
minf	6 812	0x00000138
vmhd	20	0x00000140
dinf	36	0x00000154
stbl	6 748	0x00000178
stsd	224	0x00000180
stts	3 784	0x00000260
stsc	40	0x00001128
stsz	2 492	0x00001150

Figure 6. Elecard Stream Analyzer GUI – Explorer Panel – MP4 Stream

AVI streams have descriptive headers that contain detailed information about AVI chunks, which is displayed in the panel.

☒ AVI stream	size	offset
☒ RIFF AVI	105 145 724	0x00000000
☒ LIST hdr1	4 588	0x0000000c
avih	56	0x00000018
☒ LIST str1	4 244	0x00000058
strh	56	0x00000064
strf	40	0x000000a4
JUNK	4 120	0x000000d4
☒ LIST odml	260	0x000010f4
dmlh	248	0x00001100
JUNK	3 576	0x00001200
LIST movi	105 125 844	0x00002000
idx1	11 680	0x064437dc
JUNK	628	0x06446584

Figure 7. Elecard Stream Analyzer GUI – Explorer Panel – AVI Stream

MKV streams have descriptive headers that contain detailed information about MKV elements, which is displayed in the panel.

☒ MKV stream	size	offset
EBML	19	0x00000000
Segment	103 710 779	0x00000018
SeekHead	64	0x00000024
Seek[0]	12	0x00000029
Seek[1]	12	0x00000038
Seek[2]	14	0x00000047
Seek[3]	14	0x00000058
Info	54	0x00001027
Tracks	2 188	0x00001062
TrackEntry[0]	1 105	0x00001068
Video	24	0x000014a2
TrackEntry[1]	1 077	0x000014bc
Audio	13	0x000018e5
Cluster[0]	2 624	0x000018f4
Cluster[1]	2 043	0x0000233a
Cluster[2]	687 534	0x00002b3b

Figure 8.

Elecard Stream Analyzer GUI – Explorer Panel – MKV Stream

3.2.3 Main Window Panel

This panel contains information about packets.

Figure 9.

Elecard Stream Analyzer GUI – Main Window Panel

Circles on the left side of the panel indicate type of information displayed.

The red circle stands for an error, the blue one is either a warning or a message.

Offset indicates the beginning of the packet position in the file. Types of packets are color coded. Braces show the parameters that can be chosen in the **Property** panel.

The scrollbar indicates red and blue messages found in the displayed packets. Search results are highlighted orange.

Click the packet to see corresponding data in the **Property** panel, **Hex Viewer** panel and **Messages** panel. Double left-click on the packet in the **Main Window** panel opens property data of the corresponding packet. To select the required data range, left-click or right-click and press the Shift key. To expand/collapse packet property data in Main Window Panel, use Right/Left arrow keys. Navigation through stream elements is provided using the hotkey Shift + Up/Down arrows.

3.2.4 Property Panel

This panel indicates properties of the chosen packet.

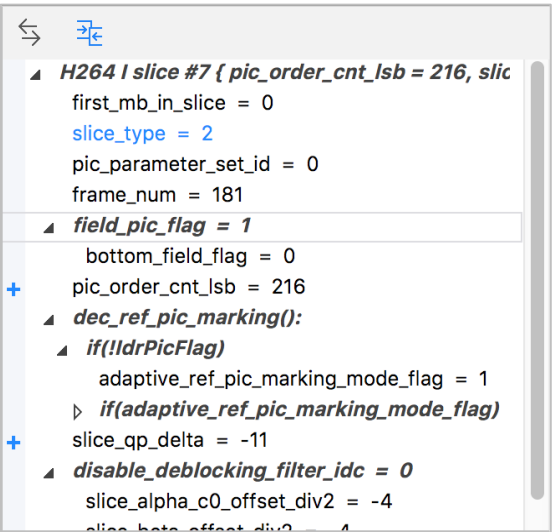


Figure 10.

Elecard Stream Analyzer GUI – Property Panel – Detailed Information about packet (from the Main Window Panel)

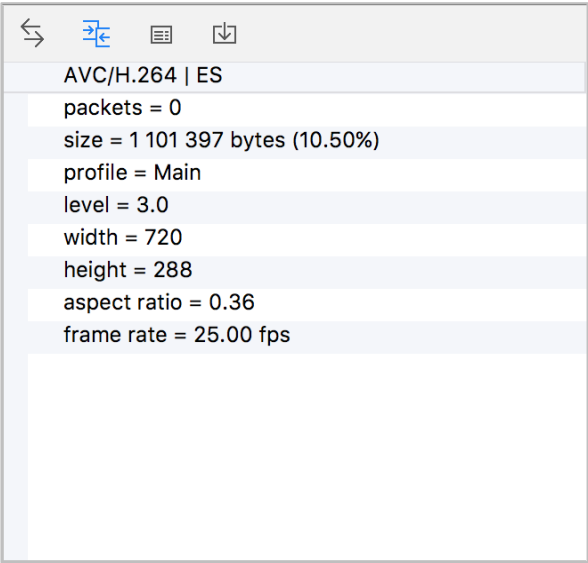


Figure 11.

Elecard Stream Analyzer GUI – Property Panel – Detailed Information about stream (from the Explorer Panel)

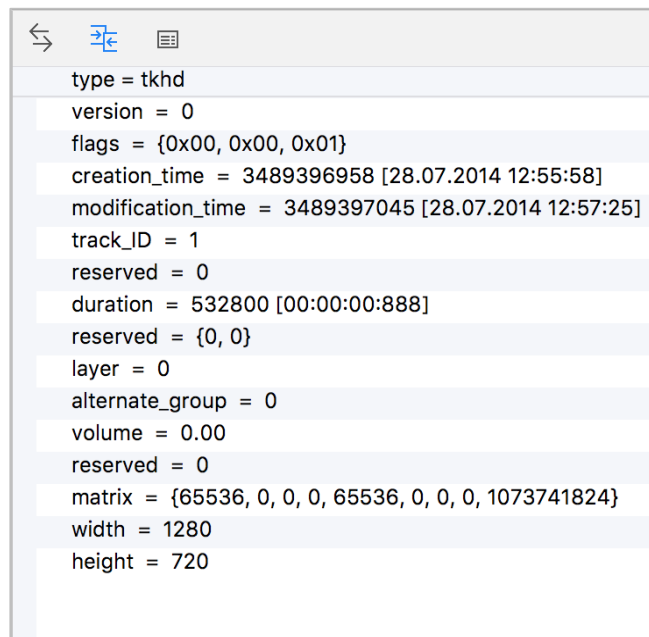


Figure 12. Elecard Stream Analyzer GUI – Property Panel – Detailed Information about chosen box (atom) (from the Explorer Panel)

To display a parameter in the Main Window panel, right-click on the necessary parameter and choose **Show in header title**. To hide a parameter, click on the  button in front of a parameter in the Property panel.

 **Compare mode** – the chosen packet will be compared with the previous chosen one. The differences in parameters will be highlighted blue.

 **Sync mode** – automatically changes data displayed in the **Property** panel without left-clicking the **Main Window** panel

 **Dump** – a function in the explorer mode for saving elementary stream.

3.2.5 Hex Viewer Panel

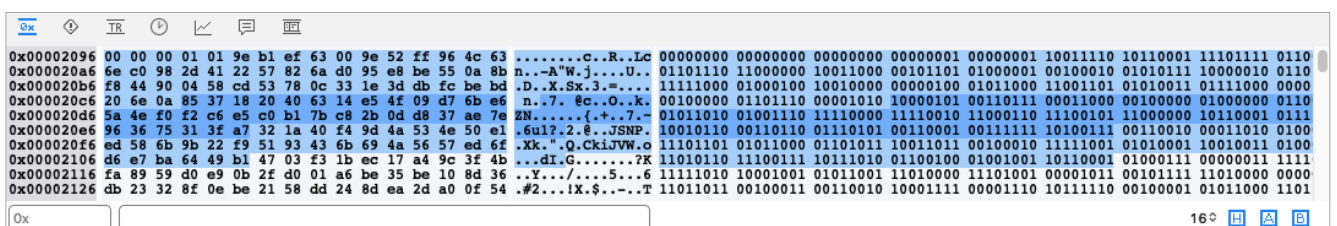


Figure 13. Elecard Stream Analyzer GUI – Hex Viewer Panel

The HEX viewer represents a file in the hex mode and allows navigation by the offset. The buttons also provide **ascii**  and **binary**  modes.

In the **Columns** drop-down list, you can choose number of bytes to show in the panel.

The Offset field  allows searching by the offset.

The **Search** text box allows searching the necessary byte sequence in the file (helps to search for start-codes).

3.2.6 Messages Panel

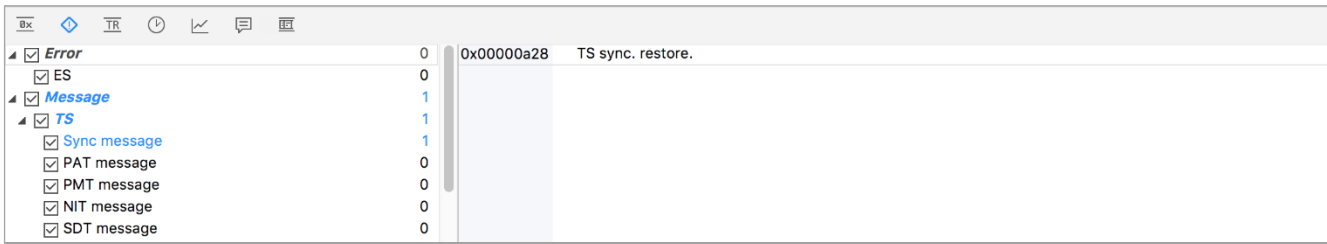


Figure 14. Elecard Stream Analyzer GUI – Messages Panel

The **Messages** panel displays a log of the stream errors analysis. The left panel is used to form the list of errors indicated in the log. It also provides a list of possible error or message types. If an error occurs, it is highlighted blue and you see the number of errors or messages found in the file. The right panel provides a list of occurred errors or messages. The red color stands for first priority errors. The blue color is the indicator of warnings and errors of second and third priority.

Double-clicking the line in the right panel sets position in **Main Window** to the packet comprising the selected error.

3.2.7 TR101-290 Panel

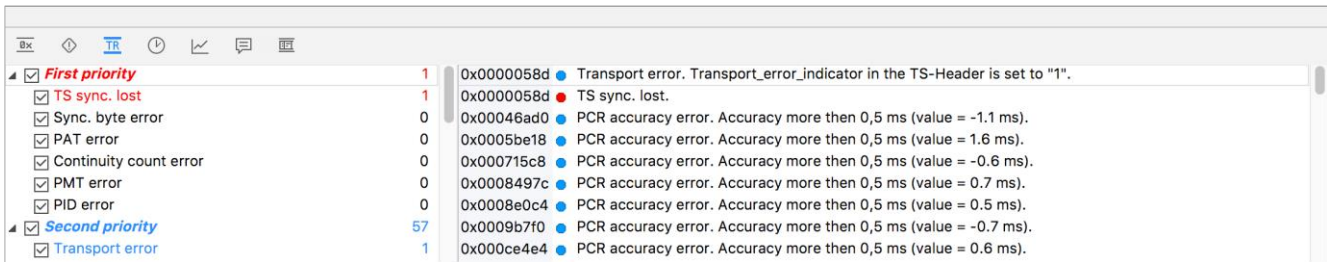


Figure 15. Elecard Stream Analyzer GUI – TR101-290 Panel

The **TR101-290** panel displays a log of the transport stream analysis. The left panel is used to form the list of errors indicated in the log. It also provides a list of possible error or message types. If an error occurs, it is highlighted blue and you see the number of errors or messages found in the file. The right panel provides a list of occurred errors or messages. The red color stands for first priority errors. The blue color is the indicator of warnings and errors of second and third priority.

Double-clicking the line in the right panel sets position in **Main Window** to the packet comprising the selected error.

If an indicator is set, then the TS is in error. However, since the indicators do not cover the entire range of possible errors, it cannot be concluded that there is no error if the indicator is not set.

If the **TS sync loss** indicator is activated then all other indicators are invalid. Each indicator is activated **only as long as** at least one of the description lines is fulfilled.

Note: In the case of indicators requiring a minimum repetition rate of sections, it means that each and every section that is present for this table should have the stated repetition rate.

Any test equipment intended for the evaluation of these parameters should report test results by means of the indicators itemized in the second column of the tables followed below. The tables describe detectable errors according to their priority types.

Table 2. First Priority Errors

Error	Description
TS sync. loss	Loss of synchronization with consideration of hysteresis parameters.
Sync. byte error	Sync_byte is not equal to 0x47
PAT error	Sections with table_id 0x00 do not occur at least every 0,5 s on PID 0x0000 Section with table_id other than 0x00 found on PID 0x0000 Scrambling_control_field is not 00 for PID 0x0000
Continuity count error	Incorrect packet order a packet occurs more than twice lost packet
PMT error	Sections with table_id 0x02, (i.e. a PMT), do not occur at least every 0,5 s on each program_map_PID which is referred to in the PAT. Scrambling_control_field is not 00 for all packets containing information of sections with table_id 0x02 (i.e. a PMT) on each program_map_PID which is referred to in the PAT.
PID error	Referred PID does not occur for a user specified period.

Table 3. Second Priority Errors

Error	Description
Transport error	Transport_error_indicator in the TS-Header is set to "1"
CRC error	CRC error occurred in CAT, PAT, PMT, NIT, EIT, BAT, SDT or TOT table
PCR repetition error	Time interval between two consecutive PCR is longer than 40 ms
PCR discontinuity indicator error	The difference between two consecutive PCR values (PCR _{i+1} – PCR _i) is outside the range of 0...100 ms without the discontinuity_indicator set
PCR accuracy error	PCR accuracy of selected program is not within ±500 ns
PTS error	PTS repetition period is longer than 700 ms
CAT error	Packets with transport_scrambling_control not 00 present, but no section with table_id = 0x01 (i.e. a CAT) present Section with table_id other than 0x01 (i.e. not a CAT) found on PID 0x0001

Table 4. Third Priority Errors

Error	Description
NIT actual error	Section with table_id other than 0x40 or 0x41 or 0x72 (i. e. not an NIT or ST) found on PID 0x0010 No section with table_id 0x40 (i.e. an NIT_actual) in PID value 0x0010 for longer than 10 s. Any two sections with table_id = 0x40 (NIT_actual) occur on PID 0x0010 within a specified value (25 ms or lower).
NIT other error	Interval between sections with the same section_number and table_id = 0x41 (NIT_other) on PID 0x0010 longer than a specified value (10s or higher).
SI repetition error	Repetition rate of SI tables is outside of the specified limits
Unreferenced PID	PID (other than PMT_PIDs, PIDs with numbers between 0x00 and 0x1F or PIDs user defined as private data streams) is not referred to by a PMT or a CAT within 0,5 s

Error	Description
SDT actual error	Sections with table_id = 0x42 (SDT, actual TS) is not present on PID 0x0011 for more than 2 s Sections with table_ids other than 0x42, 0x46, 0x4A or 0x72 found on PID 0x0011. Any two sections with table_id = 0x42 (SDT_actual) occur on PID 0x0011 within a specified value (25 ms or lower)
SDT other error	Interval between sections with the same section_number and table_id = 0x46 (SDT, other TS) on PID 0x0011 longer than a specified value (10s or higher)
EIT actual error	Section '0' with table_id = 0x4E (EIT-P, actual TS) is not present on PID 0x0012 for more than 2 s Section '1' with table_id = 0x4E (EIT-F, actual TS) is not present on PID 0x0012 for more than 2 s Sections with table_ids other than in the range 0x4E - 0x6F or 0x72 found on PID 0x0012 Any two sections with table_id = 0x4E (EIT-P/F, actual TS) occur on PID 0x0012 within a specified value (25ms or lower)
EIT other error	Interval between sections '0' with table_id = 0x4F (EIT-P, other TS) on PID 0x0012 longer than a specified value (10s or higher); Interval between sections '1' with table_id = 0x4F (EIT-F, other TS) on PID 0x0012 longer than a specified value (10s or higher).
EIT PF error	If either section ('0' or '1') of each EIT P/F subtable is present both must exist. Otherwise EIT_PF_error should be indicated
RST error	Sections with table_id other than 0x71 or 0x72 found on PID 0x0013. Any two sections with table_id = 0x71 (RST) occur on PID 0x0013 within a specified value (25 ms or lower)

3.2.8 Time Dynamics Panel

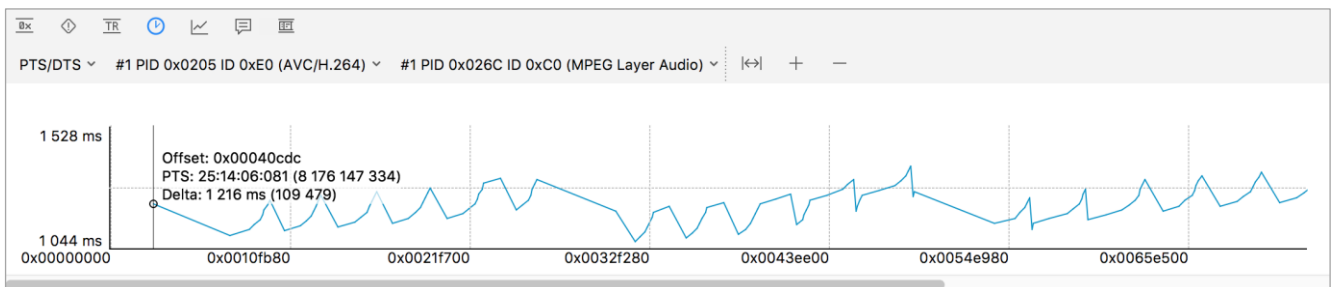


Figure 16.

Elecard Stream Analyzer GUI – Time Dynamics Panel

The **Time Dynamics** panel represents the interleaving between two streams as a diagram. To specify the streams, use the drop-down lists in the panel.

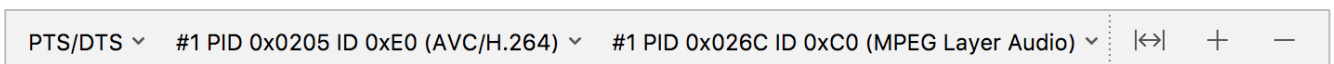


Figure 17.

Elecard Stream Analyzer GUI – Time Dynamics Panel - Panel

The chart can be built using DTSs or PTSs. If DTSs and PTSs are used together, the DTS values are used as primary ones. If DTS is absent, the PTS values are used.

These visualization options allow selecting the following Base stream characteristics for analysis (if data exists in the stream):

- **PTS/DTS Dynamics** – indicates variation of PTS/DTS with the Offset changing for the selected PID

- **PCR Dynamics** – indicates variation of PCR with the Offset changing for the selected PID
- **PCR/PTS Dynamics** – indicates variation of PTS/DTS in relation to the PCR values with the Offset changing for the selected PID
- **Offset/PCR Dynamics** – indicates variation of Offset between the PCR values for the selected PID
- **PCR Accuracy** – indicates the difference between the actual and expected PCR values for the selected PID within the range of ± 500 ns.

Moving the cursor over the window, you can see the values corresponding to the given points of the graph. Left click pins the cursor on the graph. Double-click displays the given packet in **Main Window**.

Clicking the **Fit** button  you can resize the graphic, fitting it to your screen.

Buttons  and  zoom the graphic in and out, respectively.

3.2.9 Graphics Panel

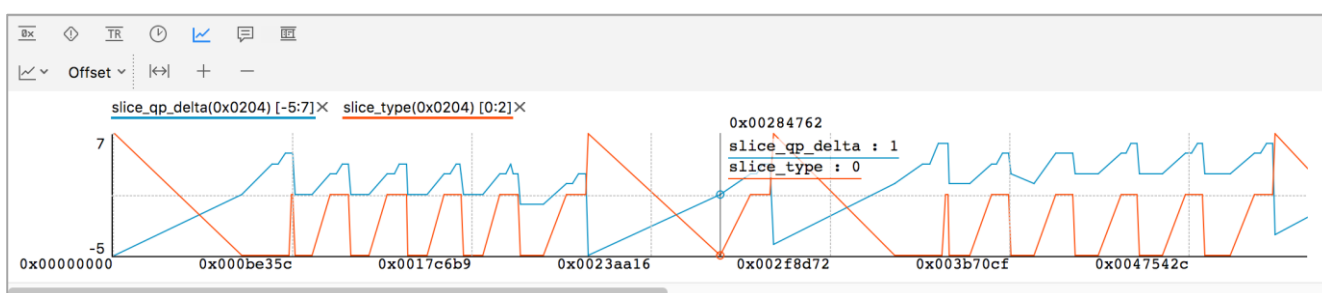


Figure 18. Elecard Stream Analyzer GUI – Graphics Panel

The **Graphics** panel serves to compare parameters of the streams.

You can add several parameters to compare. Go to the **Property** panel, right-click on the necessary parameter, choose **Add to Graphic Control**. The parameter will be added to the **Graphics** panel.

The panel allows setting up visual representation of data. The graphic can be viewed in **Line**, **Bar** and **Transition** modes. Also, you can choose ranging by **Offset** or by **Count** in the drop-down menu.

Clicking the **Fit** button  you can resize the graphic, fitting it to your screen.

Buttons  and  zoom the graphic in and out, respectively.

Stream structure – saves stream structure in the **Explorer** panel.

Full stream structure – saves only checked elements in the **Explorer** panel.

Messages – saves general info about errors.

Message details – saves all errors with their descriptions.

Visible elements – saves only opened elements in the **Main Window** panel.

Element data – saves all elements in the **Main Window** panel.

3.2.10 Comments Panel

The **Comments** panel is designed for team work: share comments on a specific frame or group of frames in the stream, navigation and switching between different instances of the same application or different applications contained in Elecard StreamEye Studio. Comments are created in the **Main**

Window panel of the application by left-clicking (or right-clicking and pressing the shift button) and selecting the required field.

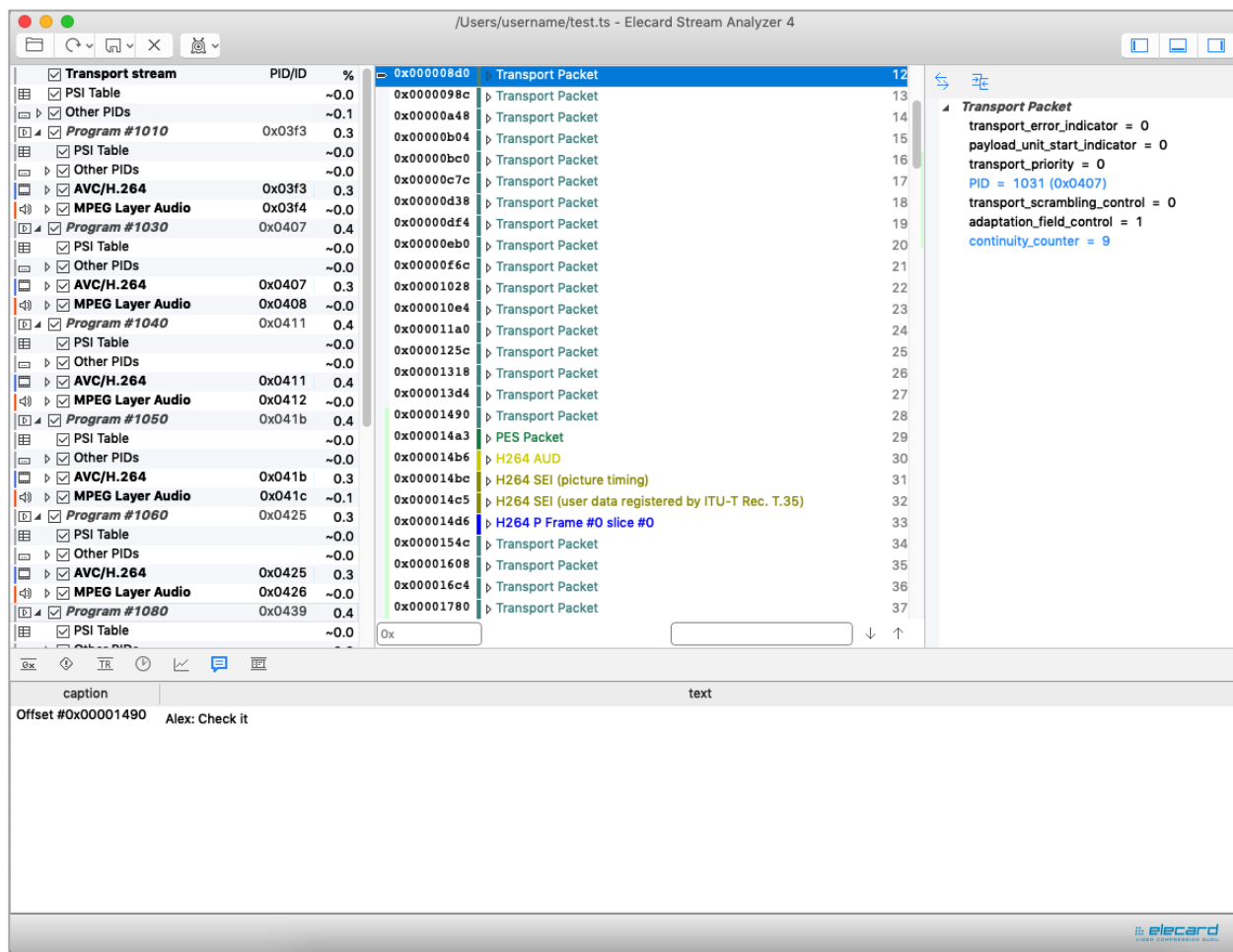


Figure 19.

Elecard Stream Analyzer GUI – Comments Panel

The **Comment** dialog should open. Enter comments in the corresponding field.

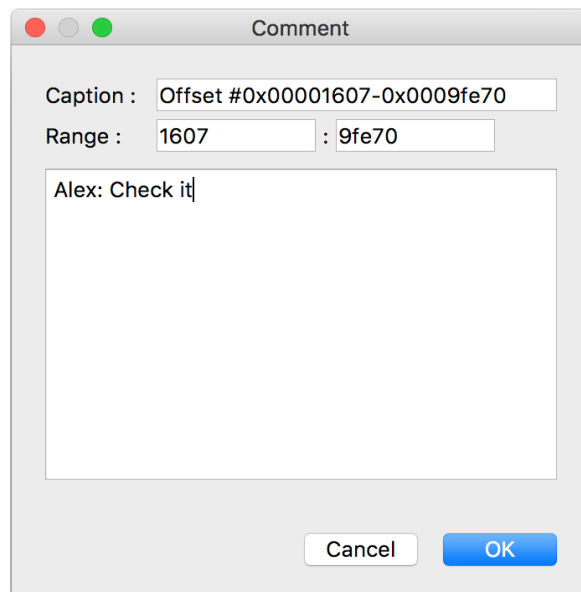


Figure 20. Elecard Stream Analyzer GUI – Comment Dialog

The comment parameters can be edited or changed before or after they are saved. To save the comment click **File ->Save Comments** and select the folder. When the comment is chosen, its position is displayed in **Main window** and highlighted green.

To edit, remove or send comments using the **Binding** mode, open the context menu by right-clicking. The **Send** option enables you to pass the selected comment to the other application of StreamEye Studio set or the other instance of Stream Analyzer. To select an application from which messages should be received or to select a type of messages to be received, go to the **Preferences tab – Binding** or see more details in the [Preferences Window](#) section.

3.2.11 EPG Panel

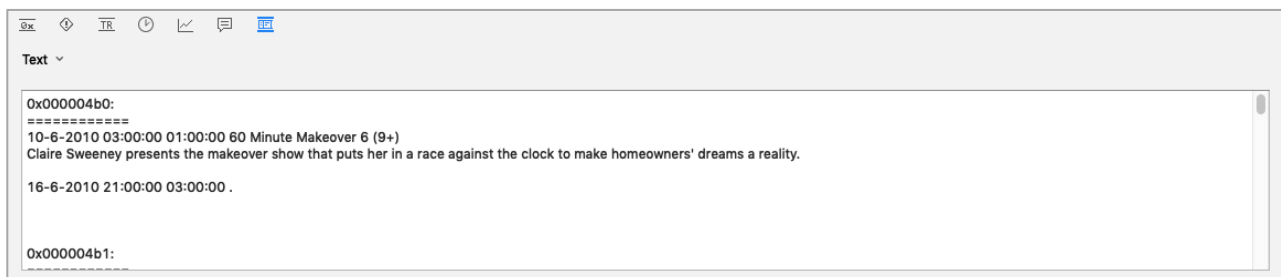


Figure 21. Elecard Stream Analyzer GUI – EPG Panel

The **EPG panel** displays scheduling information for current and upcoming broadcast programming. The **Text** mode represents programming information in a column. The **TimeLine** mode shows both information column and timeline field with time stamps. In case the successive timestamps are presented in a stream, space between them can be distributed by clicking the  and  buttons in the **TimeLine** mode. To see detailed information about the programming, hover cursor over its headline.

3.2.12 Saving Stream Information

The **Save** option allows saving different types of data: the stream itself, stream structure and information, headers, messages, errors, elements, comments etc. To select and configure the data to be saved, open the **Save** tab from the File drop-down menu or click the  button.

To save only information about the stream (without saving the stream as it is), select the **Stream Info** tab. A dialog window should open.

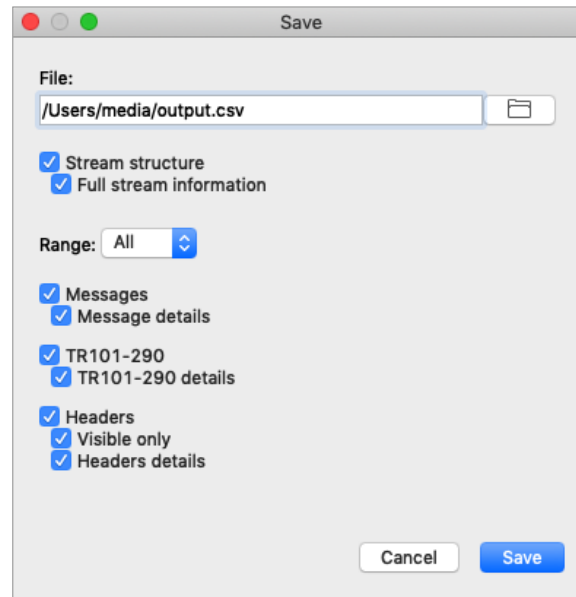


Figure 22.

Elecard Stream Analyzer GUI – Save Window – Stream Info

Stream structure – saves stream structure in the **Explorer** panel.

Full stream information – saves only checked elements in the **Explorer** panel.

Range – allows selecting a range to save data contained there:

- All – saves information on the whole stream;
- Offset – saves information on a stream portion limited by the specified offsets.

Messages – saves general information about errors.

Message details – saves errors with all details.

TR101-290 – saves general information about types of errors and their number.

TR101-290 details – saves detailed information about errors.

Headers – saves all headers without their structure.

Visible only – saves only headers selected in the **Explorer** panel and displayed in the **Main Window** panel.

Headers details – saves all headers with their structures selected in the **Explorer** panel and displayed in the **Main Window** panel.

To save a stream or its portion, select the **Dump** tab. A dialog window should open.

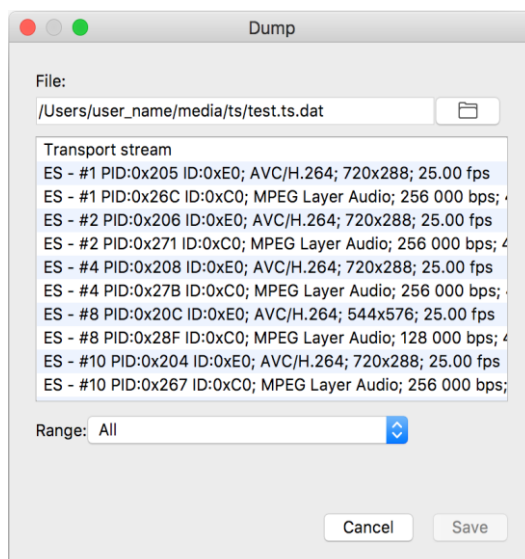


Figure 23. Elecard Stream Analyzer GUI – Save Window – Dump

Select an Elementary stream to be dumped as a whole or within a specified range.

Range – allows selecting a range to save data contained there:

- All – saves information on the whole stream;
- Offset – saves information on a stream portion limited by the specified offsets;

3.2.13 Preferences Window

To open the Preferences window, select the **StreamAnalyzer4** tab and the **Preferences** tab from the drop-down menu or click the  button. The Preferences window allows opening a file with an external application, specifying the application name and path, specifying the number of files in a list, configuring **Binding mode** settings.

Specify the number of recent files to be displayed in a list. The default value is 10 files.

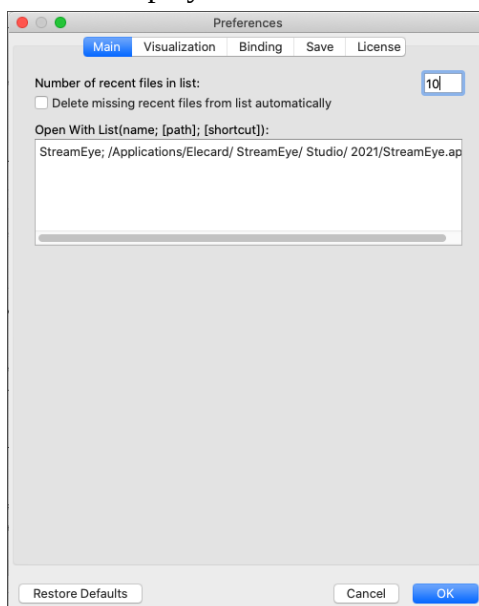


Figure 24. Elecard Stream Analyzer GUI – Preferences – Main Tab

On attempt to open a file from a list that does not exist any longer, a message notifying that the file is missing occurs. It is recommended to delete missing files from the list. To enable automatic deletion of missing files, set the checkbox for **Delete missing recent files from list automatically**.

To open the file currently opened in Stream Analyzer with another application, specify its name and path in the **Open with** field according to parameters specified there. The parameters path and shortcut are optional. If the path parameter is not specified, default application or player supporting a file format and installed in the system is opened. Each string specified in the **Open with list** field corresponds to an individual item in the menu **File – Open with**, that allows easy switching between applications or players after configuring the field.

Note: The **Restore Defaults** button allows you to reset set options to their original values. Functionality of this button is not applicable to the **License** tab.

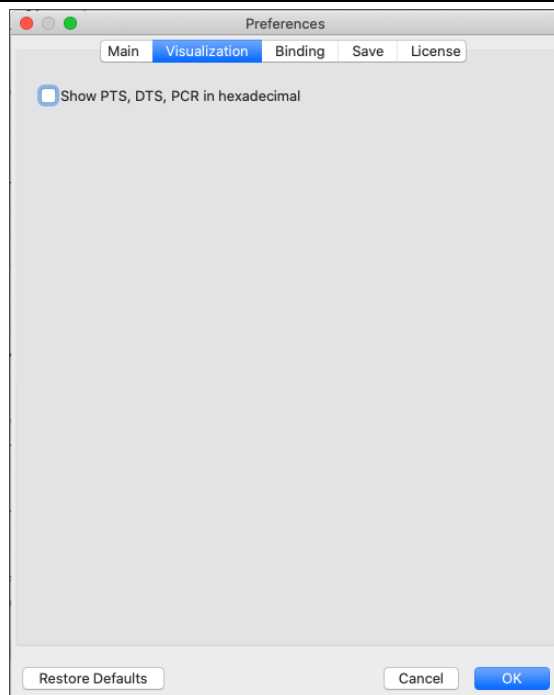


Figure 25.

Elecard Stream Analyzer GUI – Preferences – Visualization

The **Visualization** tab contains the **Show PTS, DTS, PCR in hexadecimal** checkbox that allows the representation of PTS, DTS, PCR values in the hexadecimal format.

The **Binding** tab contains settings that allow configuration of the interaction between Elecard StreamEye Studio applications. In the **Binding** Mode, exchanging information between applications is provided via a specified port. By activating checkboxes, you can select which applications should send messages and set messages type that should be communicated. Additionally, the synchronization of applications window size and switching between controls can be enabled.

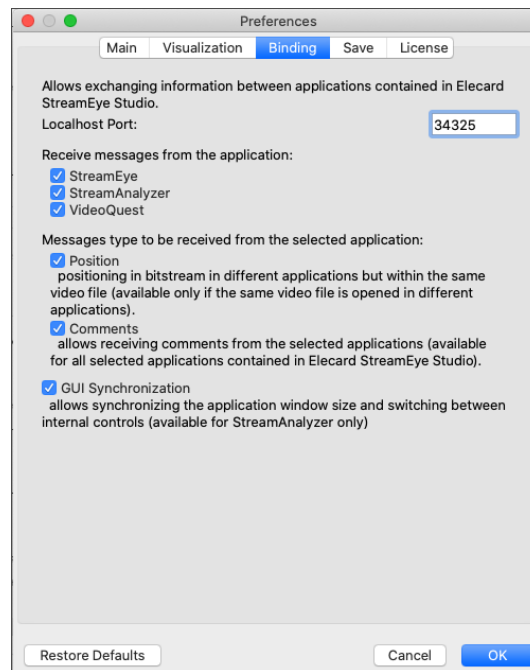


Figure 26.

Elecard Stream Analyzer GUI – Preferences – Binding Tab

Receive messages from the application – allows selecting the applications contained in Elecard StreamEye Studio from which messages are received: StreamEye, StreamAnalyzer, VideoQuest.

Messages type to be received from the selected application – allows selecting a type of messages to be received from the selected applications:

- **Position** – positioning in bitstream in different applications but within the same video file (available only if the same video file is opened in different applications).
- **Comments** – allows receiving comments from the selected applications (available for all selected applications contained in Elecard StreamEye Studio)

GUI Synchronization – allows synchronizing the application window size and switching between internal controls (available for StreamAnalyzer only).



Figure 27.

Elecard Stream Analyzer GUI – Preferences – Save Tab

The Save tab in the Preferences dialog allows saving data using the following options:

-
- **Decimal separator in .csv file** - allows selecting decimal comma or decimal dot to separate an integer part from a fractional one of a real number.
 - **Data delimiter in .csv file** - allows selecting comma or semicolon to specify boundaries in data stream.

4. Command Line Tool

Elecard Stream Analyzer functionality is available in console version now, so you can run the tool from the command line.

Stream Analyzer Command Line tool is designed for insight into MPEG-1 Video/Audio, MPEG-2 Video/Audio, AAC, Dolby Digital Audio, AVC/H.264, VC1, HEVC/H.265, VVC/H.266, AV1, AVS3, and VP9 contained in TS, PS, MP4, AVI, FLV, IVF, MKV/WebM and HEIC file formats.

Stream Analyzer Command Line tool allows automatic analysis of video arrays. You can customize a sequence of commands to perform your specific tasks and automate your routine tasks.

Get to the terminal and enter the command to start the application first and proceed with the command line options. In general, the syntax is as follows:

```
./streamanalyzer-cli config.xml /in:input.ts /out:output
```

To streamline automatic analysis of several files the following rules are introduced. They reassign input and output parameters for configuration files.

Note: You should not relocate the <product> console without all the other applications of the product contained in the installation pack, otherwise the console application will not start.

Notes on the command line rules:

- Parameters with default value are not required and can be omitted;
- Rules for input and output file names:
 - parameter /in:<file path> in the command line overwrites the value of the tag <input/> in the configuration file;
 - parameter /out:<file prefix> in the command line overwrites the value of the tag <output/> in the configuration file;
 - if <output/> tag is missing (or not specified), output file names are formed from input file name plus suffix (if the latter is specified);
 - if output file for a tag/section is not specified, the output file name for the tag/section is formed from the name specified in the <output/> tag plus suffix depending on the name of the tag/section, e.g. ".message.csv", ".header.csv", etc.

in	input file path
out	output files prefix
file path	path to the file
config path	path to the config file
sfx	suffix of all output files in config
dc	use current run directory as default path for files

4.1 Configuration File

Sample Config.xml file – contains all the commands performed by the application and reflects overall operation results:

```
<SA version="1">
<input file="...">
<output storage="union(default)/split" separator=";(default)/,"
delimiter=".(default)/," file="...">
<range start="(default: from file begin)" stop="(default: to file end)"
mode="hex(default)/dec"/>
<info>
  <stream full="on(default)/off" file="...">
  <message details="on(default)/off" file="...">
  <tr101290 details="on(default)/off" file="...">
  <headers details="on(default)/off" file="...">
    <condition parameter="name" comparison="full/substring(default)">
      <range value="min-max"/>
      <equal value="value"/>
      <less value="min"/>
      <more value="max"/>
      <not value="value"/>
    </condition>
  </headers>
</info>
</SA>
```

4.2 List of Commands

List of commands contains information on the command and all parameters which may be used within a command, and their description.

Command	Description
<input/> file	The tag is used to configure the input file; Sets the input file path.
<output/> storage separator delimiter file	The tag is used to configure the input file; Save output information into split or union storage (union (by default) / split); Defines the symbol/string to separate the output data (; (by default) / ,); Delimits the fractional part with floating point (. (by default) / .); Sets the output file path.
<info/>	The tag is used to define the type/-s of information for output.
<stream/> full file	The tag is used to output general information about the analyzed stream; Contains information on data (on (by default) / off); Sets the path for the output stream information.
<message/> details file	The tag is used to output messages that appear during analysis; Contains information on message tree data (on (by default) / off); Sets the output file path where the messages will be saved.
<tr101290/> details file	The tag is used to output TR101290 error that appear during analysis; Contains information on message TR101290 error (on (by default) / off); Sets the output file path where the messages will be saved.
<headers>	The tag is used to define the output headers information;

details	Contains full information on header data (on (by default) / off);
file	Sets the path to the output file with the headers information.

4.3 Features of Command Line Tool

- Analysis the whole array of video files automatically;
- Addressing specific and complex challenges with a few lines of commands;
- Retrieving submitted results as a text file in CSV format for easy viewing in Excel;
- Customizing a sequence of commands to perform your specific tasks;
- Accelerating and automating your regular tasks;
- Getting access to most of Stream Analyzer functionality and resources through Stream Analyzer Console.